From the derivative singlet to Rindler wavepackets
A two-point construction and its four-point contour

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Abstract

We construct spatially smeared massive fields whose Rindler midpoint correlator approaches the small-deformation derivative-singlet kernel. The construction uses the established Kontorovich–Lebedev transform and works for every positive mass. A Gaussian frequency regulator gives a uniform two-point error bound and an exact radial norm. The resulting free-field four-point function fixes what an ordinary thermal contour would predict. It differs from the microscopic tracial fourth moment, making the insertion contour, rather than a fit of decay rates, the next dictionary problem. The bra/ket-dependent fake-circle map of Lin and Stanford supplies the relevant existing framework.

1 The microscopic input and a boost normalization

Our starting observable is the conserved-subtracted derivative singlet of complex DSSYK, in the full annealed normalized trace at $\beta = \mu = 0$. With standard CAR and the Hamiltonian normalization of the preceding singlet notes, it is

$$X = (O - cB)/\sqrt{W_d}, \qquad O = \sum_i [H, c_i]^\dagger [H, c_i] - \mathbb{E}A(Q),$$
$$B = H^2 - \mathbb{E}H^2(Q), \quad c = 2\lambda \frac{1-q}{1+q}, \quad W_d = 4\mathcal{J}_s^4 \left(\lambda^{-1} + \frac{4q}{1+q}\right), \qquad q = e^{-4\lambda}. \tag{1}$$

Here A denotes the displayed sum before subtraction, and Q is fermion number. First take $N, p \to \infty$ at $p^2/N = \lambda > 0$, then take $\lambda \to 0$ at fixed string time t. The established two-point result is

$$\tau(X(t)X) \longrightarrow \operatorname{sech}^4(2\mathcal{J}_s t), \qquad \tau(X^4) = 2 + q^4 \longrightarrow 3. \tag{2}$$

All statements below use $\hbar = c_{\text{light}} = 1$. The bulk comparison is a free scalar in flat $(1+1)$-dimensional Rindler space,

$$ds^2 = -\rho^2 \mathrm{d}\eta^2 + \mathrm{d}\rho^2, \qquad \rho > 0, \qquad \beta_\eta = 2\pi. \tag{3}$$

The state is the Minkowski vacuum restricted to the right wedge. It is thermal for boosts, with local proper temperature $1/(2\pi\rho)$ [3, 1]. This is the flat-space setting proposed for the meson sector by Miyashita, Sekino and Susskind [1].

Set, as a candidate comparison of clocks,

$$\eta = 4\mathcal{J}_s t, \qquad C(\eta) = \operatorname{sech}^4(\eta/2). \tag{4}$$

This also makes the previously obtained string growth $e^{4\mathcal{J}_s t}$ equal to boost growth e^η. It corresponds to inverse temperature $\beta_t = \pi/(2\mathcal{J}_s)$ for the comparison kernel. Rahman–Susskind's cord formula has argument $\mathcal{J}_0 t_s$ and temperature $\mathcal{J}_0/\pi$ [2]; matching arguments here sets $\mathcal{J}_0 = 2\mathcal{J}_s$. This is a normalization of the two kernels, not a derivation of a particular observer radius. Proper time at fixed ρ_0 would be $\tau_{\text{prop}} = \rho_0 \eta$.

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2 Thermal completion of the singlet spectrum

Use the Fourier convention

$$C(\eta) = \int_{\mathbb{R}} \frac{\mathrm{d}\nu}{2\pi} S_C(\nu) e^{-i\nu\eta}, \qquad S_C(\nu) = \frac{8\pi}{3} \frac{\nu(1+\nu^2)}{\sinh \pi\nu}. \tag{5}$$

For example, substituting $y = e^{\eta}$ in the Fourier integral gives $16B(2+i\nu, 2-i\nu) = \frac{8}{3}|\Gamma(2+i\nu)|^2$, which is (5). In particular $S_C(0) = 8/3$ and $\int_0^\infty \nu^2 S_C(\nu)\mathrm{d}\nu/\pi = 1$.

The even spectrum has a direct interpretation as a thermal midpoint, or reflected left–right, correlator. If W is the right-wedge Wightman function, the relation and its spectral completion are

$$C(\eta) = W(\eta - i\pi), \qquad S_W(\nu) = e^{\pi\nu} S_C(\nu), \tag{6}$$

$$S_W(-\nu) = e^{-2\pi\nu} S_W(\nu), \qquad S_W(\nu) - S_W(-\nu) = \frac{16\pi}{3}\nu(1+\nu^2). \tag{7}$$

The unregulated W is a distribution; its positive-frequency spectrum grows cubically. The midpoint is an ordinary analytic function in $|\operatorname{Im}\eta| < \pi$. A trace gives an even same-operator spectrum, whereas a finite-temperature Wightman function obeys detailed balance. Thus (6) specifies which correlator is being matched. Evenness alone does not select a temperature: other spectral thermal tilts are possible. Equation (4) selects the Rindler comparison with its conventional boost period.

3 A spatial construction for every positive mass

The standard massive Rindler modes [3], specialized to two dimensions and normalized by the Klein–Gordon inner product, are

$$u_\nu(\eta, \rho) = \frac{\sqrt{\sinh \pi\nu}}{\pi} K_{i\nu}(m\rho) e^{-i\nu\eta}, \qquad \nu > 0. \tag{8}$$

Indeed the KL orthogonality relation has coefficient $\pi^2/(2\nu \sinh \pi\nu)$, while the KG product contributes 2ν. Expand $\phi = \int_0^\infty \mathrm{d}\nu (a_\nu u_\nu + a_\nu^\dagger u_\nu^*)$ with $[a_\nu, a_{\nu'}^\dagger] = \delta(\nu - \nu')$. For a real radial profile define

$$\Phi_f(\eta) = \int_0^\infty \frac{\mathrm{d}\rho}{\rho} f(\rho)\phi(\eta, \rho), \qquad F_f(\nu) = \int_0^\infty \frac{\mathrm{d}\rho}{\rho} f(\rho) K_{i\nu}(m\rho). \tag{9}$$

The Bose factors $n_\nu = (e^{2\pi\nu} - 1)^{-1}$ give

$$\begin{aligned} W_f(\eta) &= \int_0^\infty \frac{\mathrm{d}\nu}{\pi^2} \sinh(\pi\nu) |F_f(\nu)|^2 \{(n_\nu + 1)e^{-i\nu\eta} + n_\nu e^{i\nu\eta}\}, \\ C_f(\eta) &= W_f(\eta - i\pi) = \frac{1}{\pi^2} \int_0^\infty \mathrm{d}\nu \, |F_f(\nu)|^2 \cos(\nu\eta). \end{aligned} \tag{10}$$

The thermal factors have cancelled in the midpoint. We can therefore construct a profile from a prescribed positive spectrum.

Choose a frequency resolution $\epsilon > 0$ and set

$$F_\epsilon(\nu) = \sqrt{\pi S_C(\nu)}\, e^{-\epsilon\nu^2/2}, \tag{11}$$

$$f_{m,\epsilon}(\rho) = \frac{2}{\pi^2} \int_0^\infty \mathrm{d}\nu \, \nu \sinh(\pi\nu) F_\epsilon(\nu) K_{i\nu}(m\rho). \tag{12}$$

Equation (12) is the existing KL inverse, not a new transform [4, 5]. Its L^2 formulation applies: in Yakubovich's notation take the forward input to be $f_{m,\epsilon}(x/m)/x$. Then (10) becomes exactly

$$C_\epsilon(\eta) = \int_0^\infty \frac{\mathrm{d}\nu}{\pi} S_C(\nu) e^{-\epsilon\nu^2} \cos(\nu\eta) = e^{\epsilon\partial_\eta^2} C(\eta). \tag{13}$$

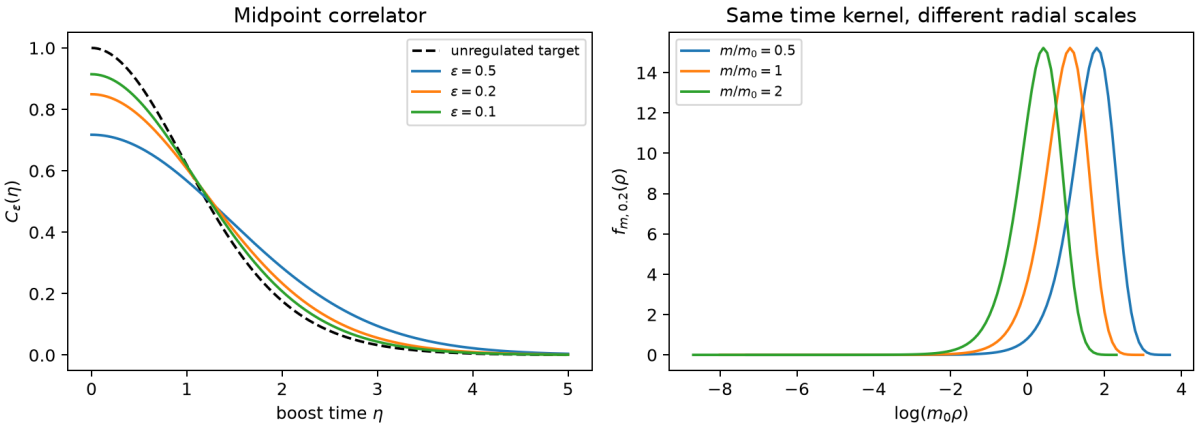


Figure 1: Left: three regulated midpoint functions approaching the singlet kernel. Right: at $\epsilon = 0.2$, masses $m/m_0 = 0.5, 1, 2$ produce different radial profiles but exactly the same temporal kernel. Profiles are smearing functions, not particle probability densities.

For all real η, not just a finite plotting interval,

$$|C_\epsilon(\eta) - C(\eta)| \le \epsilon \int_0^\infty \frac{\mathrm{d}\nu}{\pi} \nu^2 S_C(\nu) = \epsilon. \tag{14}$$

This follows from $0 \le 1 - e^{-x} \le x$. For $\epsilon < 1$, normalizing $C_\epsilon(0)$ to one changes the bound to $2\epsilon/(1-\epsilon)$. Here ϵ is a resolution parameter, independent of the microscopic deformation λ.

KL Plancherel gives an exact measure of the spatial smearing:

$$\begin{aligned} \|f_{m,\epsilon}\|^2_{L^2(\mathrm{d}\rho/\rho)} &= \frac{2}{\pi^2} \int_0^\infty \mathrm{d}\nu\, \nu \sinh(\pi\nu) |F_\epsilon(\nu)|^2 \\ &= \frac{16}{3} \int_0^\infty \mathrm{d}\nu\, (\nu^2 + \nu^4) e^{-\epsilon\nu^2} = \frac{4\sqrt{\pi}}{3\epsilon^{3/2}} + \frac{2\sqrt{\pi}}{\epsilon^{5/2}}. \end{aligned} \tag{15}$$

For each positive ϵ the ordinary Wightman variance is also finite: its integrand is proportional to $S_C(\nu) e^{-\epsilon\nu^2} \cosh \pi\nu$. These are well-defined smeared fields on the usual free-field domain. Removing the regulator preserves the midpoint function but makes the radial norm diverge. The exact unregulated kernel is not represented by an $L^2(\mathrm{d}\rho/\rho)$ profile of this form.

The mass freedom is explicit. With a reference mass m_0,

$$f_{m,\epsilon}(\rho) = f_{m_0,\epsilon}((m/m_0)\rho). \tag{16}$$

The measure $\mathrm{d}\rho/\rho$ is invariant under this dilation. Consequently neither the time kernel nor (15) fixes m. This is a concrete family of free-field realizations of our one-observable data, not a candidate meson mass spectrum. A physical stretched-horizon location, a locality constraint on f, or additional spatially resolved observables can restrict this freedom.

4 What the same bulk field predicts at four points

Having constructed a field, its ordinary thermal four-point contour is no longer adjustable independently. Write the equally spaced thermal correlator schematically as

$$F^{\mathrm{eq}}_\epsilon(\eta) = \langle \Phi_f(\eta)\Phi_f(0)\Phi_f(\eta)\Phi_f(0)\rangle_{\beta/4 \text{ gaps}}. \tag{17}$$

This denotes the analytic KMS contour with four imaginary-time gaps $\beta/4 = \pi/2$; it does not posit a trace-class density operator for an infinite continuum wedge. Wick's theorem gives the exact free answer

$$F^{\mathrm{eq}}_\epsilon(\eta) = C_\epsilon(0)^2 + 2\,\mathrm{Re}\{C_\epsilon(\eta + i\pi/2)^2\}. \tag{18}$$

The opposite pairs are separated by $\beta/2$; the adjacent pairs by $\beta/4$ and $3\beta/4$, producing the two complex-conjugate terms. As $\epsilon \downarrow 0$, the shifted arguments remain inside the midpoint strip. At zero real time,

$$C(0) = 1, \quad C(i\pi/2) = 4, \qquad \lim_{\epsilon\downarrow 0} F_\epsilon^{\mathrm{eq}}(0) = 33. \tag{19}$$

This does not equal the microscopic limit $\tau(X^4) = 3$ in (2). Rescaling the bulk field to keep its midpoint variance one does not remove the discrepancy. The specific rule "match the two-point midpoint, then identify ordinary equally spaced thermal products" therefore fails already at leading order. This calculation does not test all possible multi-insertion dictionaries.

There is a useful existing route forward. Lin–Stanford's map [6] is different for a bra and a ket insertion:

$$\phi_{\mathrm{ket}} + \pi/2 = v(\theta_{\mathrm{ket}} + \pi/2), \qquad \phi_{\mathrm{bra}} - \pi/2 = v(\theta_{\mathrm{bra}} - \pi/2). \tag{20}$$

Their small-deformation two-point inner product is proportional to $\sin^{-2\Delta}[(\phi_{\mathrm{bra}} - \phi_{\mathrm{ket}})/2]$. At $v \to 0$, the two real arcs approach opposite points of the fake circle, rather than a uniformly rescaled physical circle. Lorentzian continuation must retain v/β_B before sending $\beta_B \to 0$. This supports a midpoint comparison and directs us to their multi-particle inner products, timefolds and two-sided OPE for the four-point map. Their §4.3 already gives a leading double-commutator formula; it should be specialized with the actual insertion ordering.

Maldacena–Shenker–Stanford's Rindler example likewise distinguishes the $\beta/4$-shifted four-point function and its Minkowski ordering [7]. A scattering interpretation needs those preparation data as well as a growth exponent. In particular our known scrambling function has the representation

$$\mathcal{F}(z) = \mathbb{E}_{P,Q\sim\Gamma(4,1)} e^{-zPQ}, \qquad z = (\lambda/2)e^{\eta}. \tag{21}$$

A unitary phase e^{iaPQ} corresponds to a continuation $z = -ia$, not positive real z. The contour and external states must establish that continuation before an on-shell coupling can be assigned.

5 The next calculation

The current construction supplies explicit candidate spatial smears and a checkable thermal contour baseline. We will next specialize the Lin–Stanford/Streicher multi-point formulas to the $r = q^2$ singlet, retaining its same-species contractions, and compare their $O(\lambda)$ double commutator to the existing exact chord OTOC. This can fix the insertion prescription without guessing it from the two-point function. The string-clock short-time coefficient $1280\mathcal{J}_s^2\lambda t^2$ and the late overlap $16\lambda e^{4\mathcal{J}_s t}$ provide two separate checks.

Mass extraction then requires an independently defined radial or translation observable and a cross-correlator matrix for more than one singlet. The present mass-dilation family explains precisely why one temporal spectrum is insufficient. The conformal chord label $\Delta = 2$ alone is not a $(1+1)$-dimensional Poincare mass. Finite curvature and interacting meson scattering remain subsequent steps.

A A finite-algebra check of thermal dressing

For any finite Hermitian H, let $\tau = \mathrm{Tr}\,/d$, $Z_\beta = \tau(e^{-\beta H})$ and $\rho_\beta = e^{-\beta H}/Z_\beta$ (a density relative to τ). The fixed dressing

$$A_2 = \sqrt{Z_\beta}\, e^{\beta H/4} X e^{\beta H/4} \tag{22}$$

satisfies the exact identity $\tau(\rho_\beta^{1/2} A_2(t) \rho_\beta^{1/2} A_2) = \tau(X(t)X)$. At four points the same dressing instead gives

$$\tau\left[\prod_{j=1}^{4} \rho_\beta^{1/4} A_2(t_j)\right] = Z_\beta \tau\left[\prod_{j=1}^{4} X(t_j) e^{\beta H/4}\right]. \tag{23}$$

To cancel n equal thermal gaps in this manner requires $A_n = Z_\beta^{1/n} e^{\beta H/(2n)} X e^{\beta H/(2n)}$. Its dependence on n is why this construction is a two-point reweighting, not a single product-preserving operator map. The continuum Rindler construction in the main text uses KMS functions, not an assumption that this finite density matrix exists there.

B Numerical record and reproducibility

ϵ	$C_\epsilon(0)$	$\|f\|^2$	$F_\epsilon^{\mathrm{eq}}(0)/C_\epsilon(0)^2$
0.50	0.717380	26.7374	7.49122
0.20	0.849207	224.589	12.3802
0.10	0.914388	1195.73	17.4203
0.05	0.953942	6552.70	22.5386
0.01	0.990171	356854	30.1092

The main run performs 91 checks: Fourier integrals, heat versus spectral convolution, spectral norm, the uniform regulator bound, finite-matrix dressings, and 21 radial refinements. Radial profiles use 128 Gauss–Legendre nodes on $[0, 32]$, 25-digit Bessel evaluations, and float64 final sums; selected comparisons use 256 nodes on $[0, 40]$ and 40-digit Bessel evaluations. Their largest absolute difference is 6.6×10^{-13}. Near $\log(m\rho) = -8$ the cancellation condition reaches 4.9×10^8, so absolute agreement does not imply the same relative precision there.

An independent script uses 50-digit tanh–sinh integration at four radial points, finite-oscillator thermal traces for (18), and spatial Simpson norms on the plotted box. Numerical refinements give radial absolute differences below 5.7×10^{-13}, with relative error 1.1×10^{-6} at the smallest-amplitude point. The nine oscillator comparisons differ by at most 8.9×10^{-16}; the largest finite-box norm discrepancy is 4.1×10^{-10} relative. These refinements are empirical; no interval bound on radial quadrature or the omitted spatial tail is claimed. Equations (14) and (15) are analytic statements about the full integrals.

Code is in `code/rindler_singlet_packet.py` and the corresponding `run_` and `check_` scripts. The result directory is `results/susskind-rindler-packet/`. It records data, parameters, dependency versions and input hashes. The build script runs `pdflatex` twice; the local manifest binds source, output and rendered-page hashes. This stage is saved locally without a Git commit or push.

References

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